

**Project Design Phase-II**  
**Solution Requirements (Functional & Non-functional)**

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|----------------------|--------------|
| <b>Date</b>          | 18 June 2026 |
| <b>Project Name</b>  | HouseHunt    |
| <b>Maximum Marks</b> | 4 Marks      |

**Functional Requirements:**

Following are the functional requirements of the proposed solution.

| <b>FR No.</b> | <b>Functional Requirement (Epic)</b> | <b>Sub Requirement (Story / Sub-Task)</b>                                                                                                                                                                                                                                                                            |
|---------------|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-1          | User Registration & Authentication   | Register via email/password form<br>Register / login via Google (OAuth)<br>Register / login via phone number + OTP<br>Separate registration flows for Renter, Landlord, and Agent roles                                                                                                                              |
| FR-2          | User Profile Management              | Create and edit personal profile (name, photo, contact details).                                                                                                                                                                                                                                                     |
| FR-3          | Property Listing Management          | Landlord / agent can create a new property listing<br>Upload multiple photos and a virtual-tour video per listing<br>Set property details: type, size, amenities, rent, deposit, availability date<br>Edit, pause, or delete an existing listing<br>Mark a property as Rented / Sold to remove it from active search |
| FR-4          | Property Search & Discovery          | Search listings by city, locality, or map area<br>Filter by property type (apartment, villa, PG, commercial)<br>Filter by price range, number of bedrooms / bathrooms, and amenities<br>Sort results by relevance, price, or date posted<br>View property on an interactive map                                      |
| FR-5          | Virtual Tour & Media Viewer          | Play embedded 360° / walkthrough video tour within the listing<br>Browse full-screen image gallery with zoom                                                                                                                                                                                                         |

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|-------|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |                                         | View floor plan / layout diagram when available                                                                                                                                                                                                                                                                              |
| FR-6  | Property Details & Market Insights      | <p>Display full property details (description, amenities, nearby landmarks)</p> <p>Show average rent trends for the selected locality</p> <p>Display a price-comparison widget against similar listings nearby</p> <p>Show Walk Score, Transit Score, and nearby schools / hospitals</p>                                     |
| FR-7  | In-App Communication                    | <p>Renter can send an enquiry / message to landlord or agent from listing page</p> <p>Real-time in-app chat between renter and landlord / agent</p> <p>Schedule and manage property visit appointments within the app</p> <p>Push notifications and email alerts for new messages and visit reminders</p>                    |
| FR-8  | Rental Application & Negotiation        | <p>Renter submits a formal rental application with personal and financial details</p> <p>Landlord reviews, accepts, counter-offers, or rejects the application</p> <p>In-app negotiation thread to agree on rent, deposit, and terms</p> <p>Application status tracking (Submitted → Under Review → Accepted / Rejected)</p> |
| FR-9  | Digital Agreement & Document Management | <p>Generate a standardised rental / lease agreement from agreed terms</p> <p>E-signature support for both renter and landlord</p> <p>Securely store signed agreements and KYC documents in the user's account</p> <p>Download / share documents as PDF</p>                                                                   |
| FR-10 | Payment & Transaction Management        | <p>Pay security deposit and advance rent online via UPI, card, or net banking</p> <p>Set up recurring monthly rent reminders and payment tracking</p> <p>View full payment history and download receipts</p> <p>Landlord receives payout to linked bank account</p>                                                          |

|       |             |                                                                                                                                                                                                                |
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| FR-13 | Admin Panel | <p>Manage and moderate user accounts (suspend, ban, restore)</p> <p>View platform-wide analytics: listings, searches, applications, revenue</p> <p>Manage subscription plans, promotions, and coupon codes</p> |
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### Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

| NFR No. | Non-Functional Requirement | Description                                                                                                                                                                                                                                                                                                                         |
|---------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NFR-1   | <b>Usability</b>           | The application shall provide an intuitive, responsive UI consistent across iOS, Android, and web browsers. New users must be able to search for and enquire about a property within 3 minutes of first launch without any onboarding assistance.                                                                                   |
| NFR-2   | <b>Security</b>            | All data in transit must be encrypted using TLS 1.2 or higher. Passwords must be hashed with bcrypt (min. cost factor 12). KYC documents and agreements must be stored encrypted at rest (AES-256). The platform must pass OWASP Top-10 vulnerability checks before each major release.                                             |
| NFR-3   | <b>Reliability</b>         | The system shall maintain 99.5% uptime on a monthly basis (excluding planned maintenance windows). Automated health checks must detect and alert on service failures within 60 seconds. All critical data must be backed up daily with a Recovery Point Objective (RPO) of 24 hours and a Recovery Time Objective (RTO) of 4 hours. |
| NFR-4   | <b>Performance</b>         | Property search results must return within 2 seconds for up to 100,000 active listings. Listing detail pages (including images) must load within 3 seconds on a 4G connection. The system must handle at least 5,000 concurrent users without degradation in response time.                                                         |
| NFR-5   | <b>Availability</b>        | The platform must be accessible 24 hours a day, 7 days a week, 365 days a year. Planned maintenance must be scheduled outside peak hours (10 PM – 5 AM IST) and announced to users at least 24 hours in advance. A read-only fallback mode shall be available                                                                       |

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|-------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |                    | during maintenance so users can browse listings.                                                                                                                                                                                                                                                                                               |
| NFR-6 | <b>Scalability</b> | <p>The backend must support horizontal scaling to handle a 10× increase in traffic (e.g., during festive season demand spikes) without manual intervention.</p> <p>The database must support sharding by city/region.</p> <p>The platform architecture must allow addition of new cities or countries within 2 weeks of business decision.</p> |